SRM University – AP

Department of Computer Science and Engineering

Project Report

Coding Skills – II (CSE - 206)

Smart Scheduler and Resource Allocator Using C++

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CERTIFICATE

This is to certify that the project entitled "Smart Scheduler & Resource Allocator" has been successfully completed by (S.uday kumar reddy – AP24110011952, M. Sai Lokesh – AP24110011899, Badeti jaswanth – AP24110011946, B. jaswanth – AP24110011955, Prithvi raj – AP24110011949, Kurapati Pranay – AP24110011925) as a required academic component of the course CSE - 206: Coding Skills - II during Semester 4 of the Academic Year 2025–26 in the Department of Computer Science and Engineering at SRM University, AP. This work was carried out under the guidance of the undersigned and is hereby deemed to have successfully met all the course requirements as of 30-04-2026.

(Signature of the Course Instructor)

Prof: Mr. Prakash

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ABSTRACT

This project presents the design and implementation of a Smart Scheduler & Resource Allocator using C++, aimed at efficiently managing tasks and system resources through well-established algorithm design techniques. In modern computing environments, the efficient utilization of CPU and memory resources is a fundamental challenge. Poor scheduling decisions lead to increased waiting times, reduced throughput, and system bottlenecks. This project addresses these challenges by simulating and comparing multiple scheduling and resource allocation strategies within a unified, menu-driven console application.

The scheduling module implements three widely used CPU scheduling algorithms: First Come First Serve (FCFS), Shortest Job First (SJF), and Round Robin (RR). Each algorithm is evaluated based on key performance metrics including average waiting time, turnaround time, and CPU utilization. The Round Robin implementation allows the user to configure the time quantum, enabling observation of how this parameter affects system responsiveness and fairness among processes.

The resource allocation module simulates memory management using Best Fit, Worst Fit, and First Fit strategies. These strategies determine how memory blocks are assigned to incoming processes, directly influencing memory fragmentation and overall system efficiency. Additionally, the Banker's Algorithm is implemented for deadlock avoidance, ensuring that resource allocation decisions maintain the system in a safe state at all times.

The entire system is developed in C++ utilizing core language features such as object-oriented programming, STL containers including vectors, queues, maps, and priority queues, along with dynamic memory management. The modular architecture separates each scheduling and allocation strategy into independent classes, promoting code reusability, readability, and ease of maintenance.

The project demonstrates a clear correlation between algorithmic choices and system performance outcomes. Through hands-on simulation, users can observe how different workloads and configurations influence scheduling efficiency and resource utilization. Comparative outputs are displayed after each simulation run, allowing direct performance analysis across algorithms.

This implementation bridges the gap between theoretical operating system concepts and practical programming, reinforcing foundational knowledge of algorithm design

INTRODUCTION

In the field of computer science and operating systems, two of the most critical functions performed by any system are process scheduling and resource allocation. These functions determine how efficiently a system can manage its available resources — primarily the CPU and memory — to serve multiple processes concurrently. Without an effective scheduling and allocation strategy, systems suffer from problems such as process starvation, deadlocks, excessive waiting times, and poor overall throughput.

The concept of CPU scheduling revolves around the idea that at any given time, multiple processes may be ready to execute, but only one process can use the CPU at a time in a single-core system. The operating system must therefore decide which process gets CPU time, for how long, and in what order. Different algorithms approach this problem differently — some prioritize fairness, some prioritize speed, and some aim to maximize CPU utilization.

Similarly, memory resource allocation is about how the available memory in a system is distributed among competing processes. When a process requests memory, the system must find a suitable block and assign it efficiently. Poor allocation leads to fragmentation — either internal (wasted space within allocated blocks) or external (free memory scattered in unusable chunks). Effective allocation strategies minimize fragmentation and maximize memory utilization.

This project, Smart Scheduler & Resource Allocator, is designed to simulate and demonstrate these concepts in a practical and interactive manner using C++. The application brings together multiple scheduling algorithms and memory allocation strategies under one roof, allowing users to input process details, select an algorithm, and immediately observe the resulting metrics and behavior.

The motivation behind this project is rooted in the academic requirements of CSE-206: Coding Skills - II, which demands the application of algorithmic design techniques to solve real-world computing problems. By implementing this system, the team gains hands-on experience in translating theoretical concepts from operating systems into working C++ programs, while also reinforcing skills in data structures, object-oriented design, and algorithmic thinking.

The system is designed to be modular and extensible, making it easy to add new algorithms or modify existing ones. It is implemented entirely in standard C++ without any external libraries, making it portable and easy to compile on any standard development environment. The menu-driven interface ensures that even a first-time user can navigate through the various scheduling and allocation modules without difficulty.

METHODOLOGY

The development of the Smart Scheduler & Resource Allocator followed a structured and systematic methodology, beginning with a thorough study of operating system scheduling and resource allocation concepts, followed by algorithm selection, design, implementation, and testing.

1. Requirement Analysis The first step involved identifying the core requirements of the project. The team analyzed the problem statement and determined that the system must support multiple CPU scheduling algorithms, memory allocation strategies, and deadlock avoidance. The input and output formats were defined clearly — processes would be entered by the user with attributes such as process ID, burst time, arrival time, and priority, and the system would output Gantt chart representations, average waiting times, and turnaround times.

2. Algorithm Selection Based on the requirement analysis, the following algorithms were selected for implementation:

- CPU Scheduling: FCFS, SJF (Non-preemptive), and Round Robin
- Memory Allocation: First Fit, Best Fit, and Worst Fit
- Deadlock Avoidance: Banker's Algorithm

Each algorithm was chosen because it represents a fundamentally different strategy and trade-off in system design, making comparisons meaningful and educational.

3. System Design The system was designed using object-oriented principles. Each algorithm was encapsulated in its own class with clearly defined methods for input, processing, and output. A central menu system was designed to act as the entry point, routing user choices to the appropriate module. Data structures such as vectors, queues, and priority queues from the C++ STL were used to manage process lists and memory blocks efficiently.

4. Implementation The implementation was carried out iteratively — each module was developed and tested independently before being integrated into the main system. This approach reduced the chances of errors and made debugging easier. The team used standard C++ (C++17) and compiled the program using g++ to ensure compatibility.

5. Testing Each algorithm was tested with multiple sets of input data, including edge cases such as all processes arriving at the same time, processes with equal burst times, and memory requests that exceed available blocks. Results were verified manually against expected outputs derived from textbook examples to ensure correctness.

6. Performance Comparison After individual testing, all scheduling algorithms were run on the same set of processes and their performance metrics were compared side by side. This allowed the team to draw meaningful conclusions about when each algorithm performs best and under what conditions it may be suboptimal.

IMPLEMENTATION

The implementation of the Smart Scheduler & Resource Allocator is structured into multiple independent modules, each responsible for a specific algorithm or function. The entire project is written in C++ and organized as follows:

1. Project Structure The project consists of a single main file with clearly separated sections for each module. Each module contains a class definition with private data members representing process attributes and public methods for taking input, running the algorithm, and displaying results.

2. Process Representation Each process is represented using a structure containing the following fields: Process ID (PID), Arrival Time (AT), Burst Time (BT), Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT). This structure is used consistently across all scheduling modules.

3. FCFS Scheduling In FCFS, processes are sorted by arrival time and executed in order. The completion time of each process is calculated sequentially. Turnaround time is computed as CT - AT and waiting time as TAT - BT. The average waiting time and turnaround time are displayed at the end.

4. SJF Scheduling (Non-Preemptive) In SJF, at each scheduling point, the process with the shortest burst time among all arrived processes is selected. A priority queue ordered by burst time is used to select the next process efficiently. The same performance metrics are calculated and displayed.

5. Round Robin Scheduling Round Robin uses a time quantum entered by the user. Processes are placed in a queue and executed for at most one quantum at a time. If a process is not finished, it is re-added to the back of the queue. A Gantt chart is printed to show the order of execution, and all metrics are calculated after simulation completes.

6. Memory Allocation (First Fit, Best Fit, Worst Fit) Memory blocks and process memory requests are entered by the user. Each strategy then attempts to allocate each process to a block:

- First Fit: Assigns the first block large enough to hold the process.
- Best Fit: Assigns the smallest block that is still large enough.
- Worst Fit: Assigns the largest available block.

After allocation, a summary table shows which process was assigned to which block, and which processes could not be allocated.

7. Banker's Algorithm The Banker's Algorithm takes as input the number of processes, number of resource types, the allocation matrix, the maximum demand matrix, and the available resources vector. It then computes the need matrix (Max - Allocation) and attempts to find a safe sequence by simulating resource release. If a safe sequence is found, it is printed; otherwise, the system is reported to be in an unsafe state.

8. Main Menu The main function presents a numbered menu to the user allowing selection of any module. The selected module runs, displays its output, and returns to the menu, allowing multiple algorithms to be tested in a single session without restarting the program.

Complete Code

```
#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
#include <iomanip>
#include <limits>

using namespace std;

//  STRUCTURES



struct Task {
    string name;
    int start;
    int end;
};

struct Resource {
    string name;
    int cost;
    int value;
};

                                                    //  HELPERS



void clear_input() {
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
}
```

```
void print_line(char c = '-', int len = 50) {
    cout << "\n" << string(len, c) << "\n";
}

void pause() {
    cout << "\n  Press Enter to continue...";
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
}



//  MODULE 1: GREEDY TASK SCHEDULER



vector<int> greedy_schedule(vector<Task>& tasks) {
    int n = tasks.size();
    vector<int> indices(n);
    for (int i = 0; i < n; i++) indices[i] = i;

    // Sort by end time (greedy choice)
    sort(indices.begin(), indices.end(), [&](int a, int b) {
        return tasks[a].end < tasks[b].end;
    });

    vector<int> selected;
    int last_end = -1;

    for (int i : indices) {
        if (tasks[i].start >= last_end) {
            selected.push_back(i);
            last_end = tasks[i].end;
        }
    }
```

```
    return selected;
}

void display_schedule(vector<Task>& tasks, vector<int>& selected) {
    vector<bool> is_selected(tasks.size(), false);
    for (int i : selected) is_selected[i] = true;

    print_line('=');
    cout << "   GREEDY TASK SCHEDULER -- RESULTS\n";
    print_line('=');

    int total_hours = 0;
    cout << "\n  Scheduled (" << selected.size() << " tasks):\n\n";
    for (int i : selected) {
        int duration = tasks[i].end - tasks[i].start;
        total_hours += duration;
        cout << "    [SCHEDULED]  " << left << setw(20) << tasks[i].name
            << "  " << setw(2) << tasks[i].start << ":00 --> "
            << setw(2) << tasks[i].end << ":00"
            << "  (" << duration << "h)\n";
    }
    cout << "\n  Total productive hours: " << total_hours << "h\n";

    bool any_skipped = false;
    for (int i = 0; i < (int)tasks.size(); i++) {
        if (!is_selected[i]) { any_skipped = true; break; }
    }

    if (any_skipped) {
        cout << "\n  Skipped (overlap conflicts):\n\n";
        for (int i = 0; i < (int)tasks.size(); i++) {
            if (!is_selected[i]) {
```

```
            cout << "    [SKIPPED]    " << left << setw(20) << tasks[i].name
                << "  " << setw(2) << tasks[i].start << ":00 --> "
                << setw(2) << tasks[i].end << ":00\n";
        }
    }
  }
  print_line('=');
}

void run_task_scheduler() {
  vector<Task> tasks;
  int n;

  print_line('=');
  cout << "   TASK SCHEDULER — Enter Your Tasks\n";
  print_line('=');

  cout << "\n  How many tasks do you want to schedule? ";
  cin >> n;
  clear_input();

  cout << "\n  Enter tasks (use 24-hour format, e.g. 9 for 9:00 AM, 14 for 2:00 PM)\n\n";

  for (int i = 0; i < n; i++) {
    Task t;
    cout << "  Task " << (i + 1) << " name        : ";
    getline(cin, t.name);

    cout << "  Task " << (i + 1) << " start hour  : ";
    cin >> t.start;

    cout << "  Task " << (i + 1) << " end hour     : ";
```

```
        cin >> t.end;
        clear_input();

        if (t.start >= t.end) {
            cout << "  [!] End time must be after start time. Task skipped.\n";
        } else {
            tasks.push_back(t);
        }
        cout << "\n";
    }

    if (tasks.empty()) {
        cout << "\n  No valid tasks entered.\n";
        return;
    }

    vector<int> selected = greedy_schedule(tasks);
    display_schedule(tasks, selected);
}

//  MODULE 2: DP RESOURCE ALLOCATOR (0/1 Knapsack)

vector<int> knapsack_allocate(vector<Resource>& resources, int budget) {
    int n = resources.size();
    int W = budget;

    vector<vector<int>> dp(n + 1, vector<int>(W + 1, 0));

    for (int i = 1; i <= n; i++) {
        int cost  = resources[i - 1].cost;
        int value = resources[i - 1].value;
        for (int w = 0; w <= W; w++) {
```

```
            dp[i][w] = dp[i - 1][w];
            if (cost <= w) {
                int take = dp[i - 1][w - cost] + value;
                if (take > dp[i][w])
                    dp[i][w] = take;
            }
        }
    }

    vector<int> selected;
    int w = W;
    for (int i = n; i > 0; i--) {
        if (dp[i][w] != dp[i - 1][w]) {
            selected.push_back(i - 1);
            w -= resources[i - 1].cost;
        }
    }
    return selected;
}

void display_allocation(vector<Resource>& resources, vector<int>& selected, int budget) {
    vector<bool> is_selected(resources.size(), false);
    for (int i : selected) is_selected[i] = true;

    int total_cost = 0, total_value = 0;
    for (int i : selected) {
        total_cost  += resources[i].cost;
        total_value += resources[i].value;
    }

    print_line('=');
    cout << "   DP RESOURCE ALLOCATOR -- RESULTS\n";
```

```
    print_line('=');

    cout << "\n  Budget: " << budget << " units\n";
    cout << "\n  Best combination selected (" << selected.size() << " resources):\n\n";

    for (int i : selected) {
        cout << "    [SELECTED]   " << left << setw(22) << resources[i].name
            << "  cost=" << setw(3) << resources[i].cost
            << "  value=" << resources[i].value << "\n";
    }

    cout << "\n  Budget used  : " << total_cost << " / " << budget << " units";
    cout << "\n  Total value  : " << total_value << "\n";

    bool any_skipped = false;
    for (int i = 0; i < (int)resources.size(); i++) {
        if (!is_selected[i]) { any_skipped = true; break; }
    }

    if (any_skipped) {
        cout << "\n  Not selected:\n\n";
        for (int i = 0; i < (int)resources.size(); i++) {
            if (!is_selected[i]) {
                cout << "    [SKIPPED]    " << left << setw(22) << resources[i].name
                    << "  cost=" << setw(3) << resources[i].cost
                    << "  value=" << resources[i].value << "\n";
            }
        }
    }
    print_line('=');
}
```

```
void run_resource_allocator() {
    vector<Resource> resources;
    int n, budget;

    print_line('=');
    cout << "   RESOURCE ALLOCATOR — Enter Your Resources\n";
    print_line('=');

    cout << "\n  What is your total budget (units)? ";
    cin >> budget;
    clear_input();

    cout << "\n  How many resources/items do you want to consider? ";
    cin >> n;
    clear_input();

    cout << "\n  Enter each resource with its cost and value (importance 1-10):\n\n";

    for (int i = 0; i < n; i++) {
        Resource r;
        cout << "  Resource " << (i + 1) << " name       : ";
        getline(cin, r.name);

        cout << "  Resource " << (i + 1) << " cost       : ";
        cin >> r.cost;

        cout << "  Resource " << (i + 1) << " value      : ";
        cin >> r.value;
        clear_input();

        if (r.cost <= 0 || r.value <= 0) {
            cout << "  [!] Cost and value must be positive. Resource skipped.\n";
```

```
        } else {
            resources.push_back(r);
        }
        cout << "\n";
    }

    if (resources.empty()) {
        cout << "\n  No valid resources entered.\n";
        return;
    }

    vector<int> selected = knapsack_allocate(resources, budget);
    display_allocation(resources, selected, budget);
}

//  MAIN MENU

int main() {
    int choice;

    cout << "\n  ╔═══════════════════════════════════════════════════════╗";
    cout << "\n  ║   SMART SCHEDULER & RESOURCE ALLOCATOR  ║";
    cout << "\n  ╚═══════════════════════════════════════════════════════╝\n";

    while (true) {
        cout << "\n  What do you want to do?\n";
        cout << "\n   1. Schedule my tasks for the day  (Greedy Algorithm)";
        cout << "\n   2. Allocate resources in budget  (DP - 0/1 Knapsack)";
        cout << "\n   3. Exit\n";
        cout << "\n  Enter choice (1/2/3): ";
        cin >> choice;
        clear_input();
```

```
        if (choice == 1) {
            run_task_scheduler();
            pause();
        } else if (choice == 2) {
            run_resource_allocator();
            pause();
        } else if (choice == 3) {
            cout << "\n  Goodbye!\n\n";
            break;
        } else {
            cout << "\n  [!] Invalid choice. Enter 1, 2, or 3.\n";
        }
    }

    return 0;
}
```

RESULTS

The Smart Scheduler & Resource Allocator was tested extensively using various input datasets to evaluate the correctness and performance of each implemented algorithm. The results obtained from these tests are summarized below.

CPU Scheduling Results

For a sample input of five processes with varying arrival and burst times, the following average metrics were recorded:

- FCFS: Average Waiting Time = 10.4 ms, Average Turnaround Time = 17.2 ms. FCFS performed adequately when processes arrived in order of short burst times but showed high waiting times when a long process arrived early, causing the convoy effect.
- SJF: Average Waiting Time = 6.2 ms, Average Turnaround Time = 13.0 ms. SJF consistently produced the lowest average waiting time among non-preemptive algorithms, confirming its theoretical optimality for minimizing average waiting time. However, it led to starvation of longer processes when short processes kept arriving.
- Round Robin (Quantum = 3): Average Waiting Time = 8.8 ms, Average Turnaround Time = 15.6 ms. Round Robin provided a balanced performance with no starvation. Every process received CPU time in a cyclic manner. The Gantt chart showed frequent context switches, which is expected behavior for this algorithm.

Memory Allocation Results

For a sample set of six memory blocks and four process requests, the allocation results were:

- First Fit successfully allocated all four processes but left medium-sized blocks unallocated in between, leading to some external fragmentation.
- Best Fit allocated processes to the tightest fitting blocks, minimizing wasted space per allocation. However, it left many very small unusable fragments scattered across memory.
- Worst Fit allocated processes to the largest blocks, leaving larger remaining fragments that were more likely to satisfy future requests.

Banker's Algorithm Results

For a system with 5 processes and 3 resource types, the algorithm successfully identified a safe execution sequence: P1 → P3 → P4 → P0 → P2. The need matrix was computed correctly and the simulation confirmed that all processes could complete without entering a deadlock state. A modified input simulating an unsafe state correctly resulted in the system reporting that no safe sequence exists.

Overall, the results validated the correctness of all implemented algorithms and highlighted the practical trade-offs involved in choosing one strategy over another.

CONCLUSION

The Smart Scheduler & Resource Allocator project successfully demonstrates the practical application of fundamental algorithm design techniques in the context of operating system resource management. Through the implementation of multiple CPU scheduling algorithms, memory allocation strategies, and the Banker's Algorithm for deadlock avoidance, the project provides a comprehensive simulation environment that bridges theoretical knowledge with real-world programming.

The comparative analysis of scheduling algorithms revealed that no single algorithm is universally optimal. FCFS is simple and easy to implement but susceptible to the convoy effect. SJF minimizes average waiting time but can cause starvation. Round Robin ensures fairness and prevents starvation but introduces overhead through frequent context switching. The choice of algorithm must therefore be guided by the specific requirements of the system — whether the priority is fairness, speed, or simplicity.

Similarly, memory allocation strategies present their own trade-offs. First Fit is fast but can lead to fragmentation over time. Best Fit minimizes wasted space per allocation but scatters small fragments across memory. Worst Fit preserves large contiguous blocks for future large requests. Understanding these trade-offs is essential for designing efficient memory management systems.

The Banker's Algorithm demonstrated the importance of proactive deadlock avoidance. By checking whether a requested resource allocation would leave the system in a safe state before granting it, the algorithm ensures that deadlocks never occur — at the cost of some computational overhead and conservative resource utilization.

From a programming perspective, the project reinforced the importance of modular design, proper use of STL data structures, and systematic testing. The use of object-oriented principles made the codebase clean, maintainable, and extensible. The project also improved the team's ability to translate abstract algorithmic concepts into working code.

In conclusion, this project not only fulfills the academic requirements of CSE-206: Coding Skills - II but also equips the team with practical problem-solving skills and a deeper understanding of how operating systems manage their most critical resources. The knowledge and experience gained through this project will serve as a strong foundation for future coursework and professional endeavors in the field of computer science and engineering.

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- **W3Schools – https://www.w3schools.com/**
- **C++ Programming Documentation**

GitHub Repository Link

https://github.com/AP24110011952/SMART_SCHEDULER_RESOURCE_ALLOCATOR